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MTECH [AISP] Subject: DIVP

Assignment No: 3

Q.1 Consider the images shown. The image on the right was obtained by:
a) multiplying the image on the left by $(-1)^{x}$
b) computing the DFT
c) taking the complex conjugate of the transform
d) computing the inverse DFT
e) multiplying the real part of the result by $(-1)^{x+y}$. Explain (mathematically) why the image on the right appears as it does.

Ans: Let the original image be

$$f(x, y)$$

of size $M \times N$

let its 2-DFT be $f(u, v)$

where

$$f(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

and inverse DFT is

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Assume that, M & N are even

a) Multiplication by $(-1)^{x+y}$

$$g(x,y) = f(x,y)(-1)^{x+y}$$

$$(-1)^x = e^{j\pi x}$$
$$(-1)^y = e^{j\pi y}$$
$$(-1)^{x+y} = e^{j\pi(x+y)}$$

$$g(x,y) = f(x,y)\, e^{j\pi x} e^{j\pi y}$$

Using the frequency-shifting property

$$G(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

Thus, multiplying the image by $(-1)^{x+y}$ shifts its Fourier spectrum by

$$\left(\frac{M}{2}, \frac{N}{2}\right)$$

b) Computing the DFT

Now compute the DFT of $g(x,y)$

$$G(u,v) = F\{g(x,y)\}$$

From the above,

$$G(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

The spectrum has been shifted by

$$\left(\frac{M}{2}, \frac{N}{2}\right)$$

c) Taking the complex conjugate

$$H(u,v) = G^*(u,v)$$

important DFT property is

$$F^{-1}\{F^*(u,v)\} = f^*(-x,-y)$$

where the coordinates are interpreted modulo M and N

Therefore, when we take the complex conjugate of the transform and then perform the inverse DFT, the spatial image becomes reflected about both axes

Since the original image is a real image

$$f^*(x,y) = f(x,y)$$
$$f^*(-x,-y) = f(-x,-y)$$

So image undergoes a 180° rotation

d) Computing the Inverse DFT

We now calculate

$$h(x,y) = F^{-1}\{G^*(u,v)\}$$

Using the conjugate property

$$h(x,y) = g^*(-x,-y)$$

Because the image is real

$$g^*(-x,-y) = g(-x,-y)$$

Recall that

$$g(x,y) = (-1)^{x+y} f(x,y)$$

Therefore

$$h(x,y) = (-1)^{-x-y} f(x,-y)$$

$$(-1)^{-x-y} = (-1)^{x+y}$$

we get,

$$h(x,y) = (-1)^{x+y} f(-x,-y)$$

e) Multiplying the Real Part by $(-1)^{x+y}$

$$r(x,y) = (-1)^{x+y} h(x,y)$$

Substitute the expression for $h(x,y)$

$$r(x,y) = (-1)^{x+y}\left[(-1)^{x+y} f(-x,-y)\right]$$

$$r(x,y) = (-1)^{2x+2y} f(-x-y)$$

$$(-1)^{2x+2y} = 1$$

we obtain

$$r(x,y) = f(-x,-y)$$

* Properties of DFT are used

1. Frequency Shifting / centering Property

$$g(x,y) = f(x,y)(-1)^{x+y}$$

then DFT becomes

$$G(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

2 Complex Conjugation Property

$$f(x,y) \leftrightarrow F(u,v)$$

$$f^*(u,v) \leftrightarrow f^*(-x,-y)$$
$$F^{-1}\{F^*(u,v)\} = f^*(-x,-y)$$

3 Real-Image property

$$f^*(x,y) = f(x,y)$$

$$f^*(-x,-y) = f(-x,-y)$$

4. Multiplication Property / frequency Shifting Again

$$(-1)^{x+y}(-1)^{x+y} = (-1)^{2x+2y} = 1$$

$$f(x,y) \longrightarrow f(-x,-y)$$

which is 180° rotation.

Q.2 Consider a 3×3 spatial kernel that average the four closest neighbors of a point (x,y) but excludes the point itself from the average

a) Find the equivalent filter transfer function H(u,v) in frequency domain

b) Show that your result is a lowpass filter transfer function

Ans Four closest neighbors of pixel (x,y)

Top $(x-1,y)$

Bottom $(x+1,y)$

Left $(x,y-1)$

Right $(x,y+1)$

3 x 3 kernel

$$h(x,y) = \frac{1}{4}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

a) For a spatial filter, the frequency response is obtained by taking the fourier transform of kernel

The four non zero term are located at

$(-1,0), (1,0), (0,-1), (0,1)$

Using the 2D fourier transform

$$H(u,v) = \sum_x \sum_y h(x,y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Substitute the four non-zero values

1 $(x,y) = (-1,0)$

$$h(x,y) = h(-1,0) = \frac{1}{4} e^{-j2\pi\left(\frac{-u}{M}\right)}$$

$$h(-1,0) = \frac{1}{4} e^{j2\pi u/M}$$

2 $(x,y) = (1,0)$

$$h(1,0) = \frac{1}{4} e^{-j2\pi u/M}$$

3. $(x, y) = (0, -1)$

$$h(0,-1) = \frac{1}{4} e^{j2\pi v/N}$$

4. $(x, y) = (0, 1)$

$$h(0,1) = \frac{1}{4} e^{-j2\pi v/N}$$

$$H(u,v) = \frac{1}{4}\left[e^{j2\pi u/M} + e^{-j2\pi u/M} + e^{j2\pi v/N} + e^{-j2\pi v/N} \right]$$

By Eulers Identity

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta$$

We get,

$$H(u,v) = \frac{1}{4}\left[2\cos\left(\frac{2\pi u}{M}\right) + 2\cos\left(\frac{2\pi v}{N}\right) \right]$$

$$\therefore H(u,v) = \frac{1}{2}\left[\cos\left(\frac{2\pi u}{M}\right) + \cos\left(\frac{2\pi v}{N}\right) \right]$$

b) i) Response at zero frequency

$$u = 0, \quad v = 0$$

$$H(0,0) = \frac{1}{2}\left[\cos(0) + \cos(0)\right]$$

$$\cos(0) = 1, \quad H(0,0) = \frac{1}{2}(1+1) = 1$$

$$\therefore H(0,0) = 1$$

This means the DC component and very low frequency components are passed strongly

2) Behavior at Low Frequencies

$$\cos\theta \approx 1 - \frac{\theta^2}{2}$$

$$\cos\left(\frac{2\pi u}{M}\right) \approx 1$$

$$\cos\left(\frac{2\pi v}{N}\right) \approx 1$$

$$H(u,v) \approx 1$$

near the origin

Therefore, low-frequency components are passed with high gain

3) Effect on Rapid Image variations

High spatial frequencies correspond to rapid changes in intensity, such as:

- sharp edges
- fine details
- ~~averaging~~ alternating pixel pattern

The averaging operation reduces many such rapid variations because each pixel is replaced by an average of its neighbors

Thus, the filter performs a smoothing operation

Smoothing is associated with low-pass filtering

Therefore,

Four-neighbor averaging ⇒ smoothing ⇒ Low pass frequency emphasis